

From voice to Text - using spring AI for transcription

What Can Spring AI Do for Voice?

01

Transcription is the process by which text is produced from spoken audio.

- Use `OpenAiAudioTranscriptionModel` when OpenAI used
- Feed it an audio file (MP3/WAV)
- Get back text like magic

02

Various styles of Voices/Speeches can be generated from text

- Use `SpeechModel`
- Give it a message
- Get a talking byte array 🗣️

Code Sample for Transcribe

eazy
bytes

```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel transcriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel) {
        this.transcriptionModel = transcriptionModel;
    }

    @GetMapping("/transcribe-options")
    String transcribeOptions(@Value("classpath:SpringAI.mp3") Resource audioFile) {
        var transcriptionResponse = transcriptionModel.call(
            new AudioTranscriptionPrompt(audioFile,
                OpenAiAudioTranscriptionOptions.builder()
                    .language("en")
                    .prompt("Talking about Spring AI")
                    .temperature(0f)
                    .responseFormat(OpenAiAudioApi.TranscriptResponseFormat.VTT)
                    .build()));
        return transcriptionResponse.getResult().getOutput();
    }
}
```



add these dependencies

```
pom.xml (springai) x
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
31 <spring-ai.version>1.0.0</spring-ai.version>
32 </properties>
33 <dependencies>
34 <dependency>
35 <groupId>org.springframework.boot</groupId>
36 <artifactId>spring-boot-starter-web</artifactId>
37 </dependency>
38 <dependency>
39 <groupId>org.springframework.ai</groupId>
40 <artifactId>spring-ai-starter-model-openai</artifactId>
41 </dependency>
42
43 <groupId>org.springframework.boot</groupId>
44 <artifactId>spring-boot-devtools</artifactId>
45 <scope>runtime</scope>
46 <optional>true</optional>
47 </dependency>
48 <dependency>
49
pom.xml (springai) application.properties x
1 spring.application.name=springai
2 logging.pattern.console=%green(%d{HH:mm:ss.SSS}) %blue(%-5level) %red([%thread]) %yellow(%l)
3 logging.level.org.springframework.ai.chat.client.advisor=DEBUG
4 spring.ai.openai.api-key=${OPENAI_API_KEY}
```

this is the class from the package that will have, and we need to use this class

```
gai) application.properties AudioController.java OpenAiAudioTranscriptionModel.java x
OpenAI audio transcription client implementation
for backed by OpenAiAudioApi. You provide as
input the audio file you want to transcribe and the
desired output file format of the transcription of
the audio.
Since: 0.8.1
See Also: OpenAiAudioApi
Author: Michael Lavelle, Christian Tzolov,
Thomas Vitale
48 public class OpenAiAudioTranscriptionModel implements Model<AudioTranscriptionPrompt, Aud
49
50 private final Logger logger = LoggerFactory.getLogger(getClass());
51
52 private final OpenAiAudioTranscriptionOptions defaultOptions;
53
54 private final RetryTemplate retryTemplate;
55
56 private final OpenAiAudioApi audioApi;
```

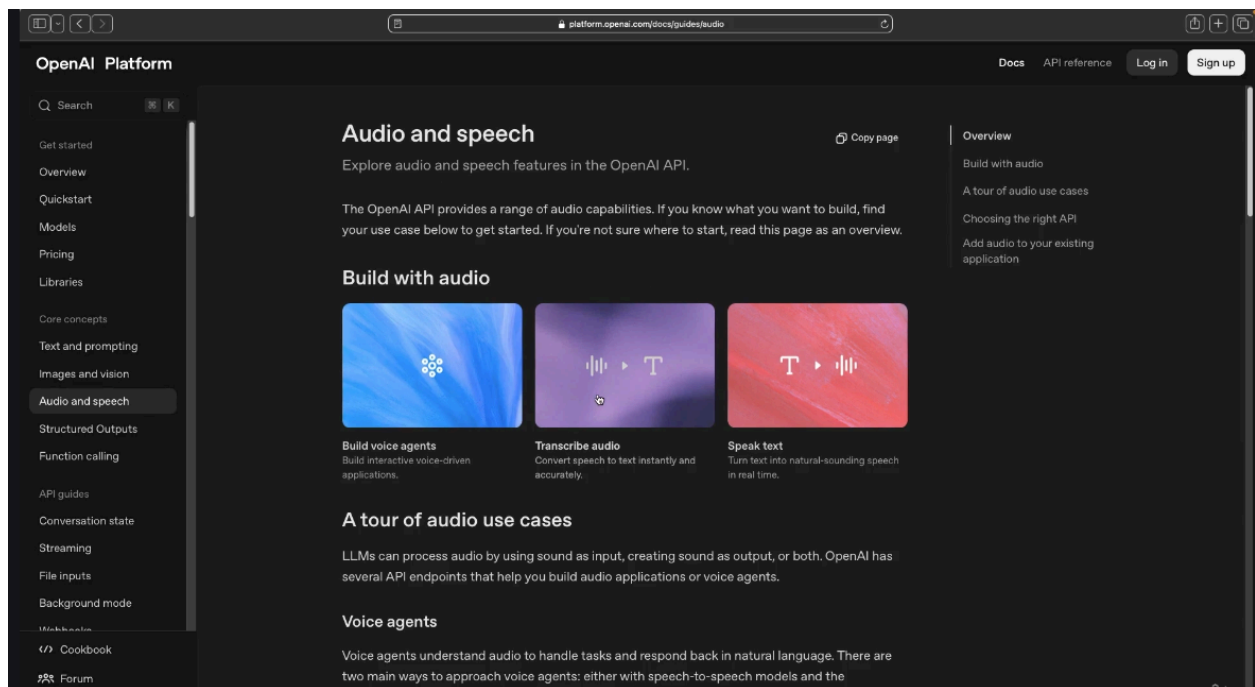
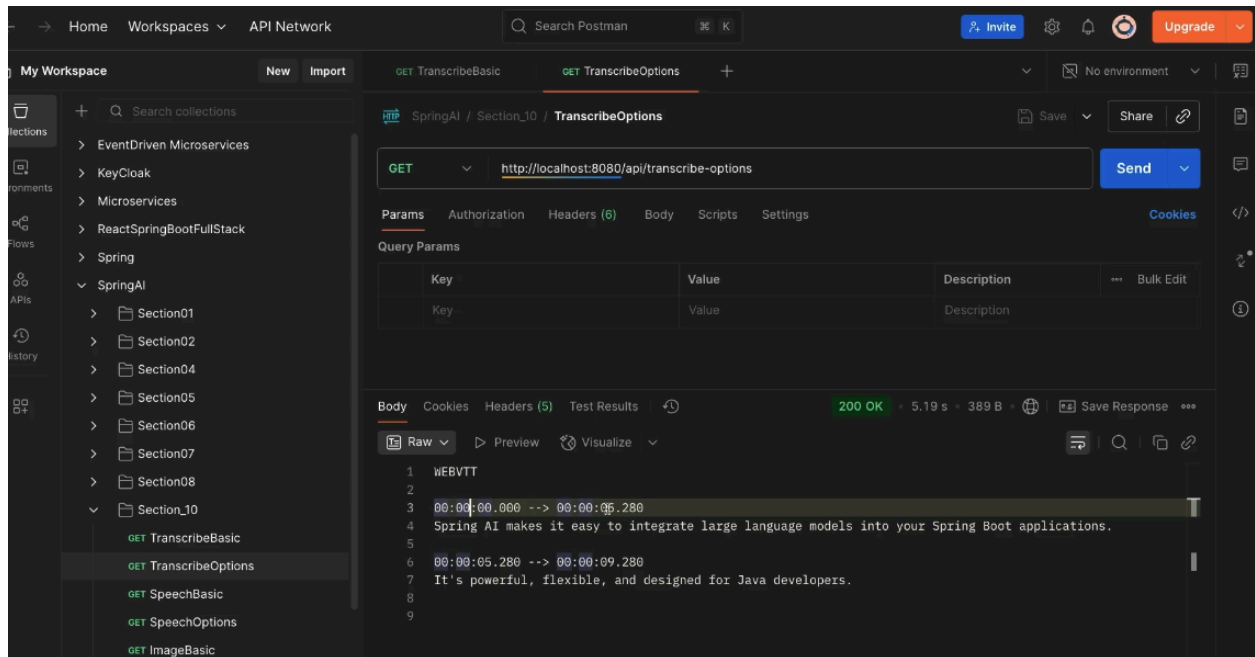
code changes in the controller

```
pom.xml (springai)  application.properties  AudioController.java x
10  @RestController
11  @RequestMapping("/api")
12  public class AudioController {
13
14      private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;
15
16      AudioController(OpenAiAudioTranscriptionModel transcriptionModel) {
17          this.openAiAudioTranscriptionModel = transcriptionModel;
18      }
19
20      @GetMapping("/transcribe")
21      String transcribe(@Value("classpath:SpringAI.mp3") Resource audioFile) {
22          return openAiAudioTranscriptionModel.call(audioFile);
23      }
24
25
26
27  }
```

part 2

if you want to enhance with any options, you use the below method

```
gai)  application.properties  AudioController.java x  OpenAiAudioTranscriptionModel.java
15  public class AudioController {
24  String transcribe(@Value("classpath:SpringAI.mp3") Resource audioFile) {
25      return openAiAudioTranscriptionModel.call(audioFile);
26  }
27
28  @GetMapping("/transcribe-options")
29  String transcribeWithOptions(@Value("classpath:SpringAI.mp3") Resource audioFile) {
30      var audioTranscriptionResponse = openAiAudioTranscriptionModel.call(new AudioTrans
31          audioFile, OpenAiAudioTranscriptionOptions.builder()
32              .prompt("Talking about Spring AI")
33              .language("en")
34              .temperature(0.5f)
35              .responseFormat(OpenAiAudioApi.TranscriptResponseFormat.VTT).build());
36      return audioTranscriptionResponse.getResult().getOutput();
37  }
38
39  }
```



From text to speech - Generating voices with speech model

Code Sample for Text to Speech

eazy
bytes



```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final SpeechModel speechModel;

    AudioController(SpeechModel speechModel) {
        this.speechModel = speechModel;
    }

    @GetMapping("/speech-options")
    String speechProviderOptions(@RequestParam("message") String message) throws IOException {
        var speechResponse = speechModel.call(new SpeechPrompt(message,
            OpenAiAudioSpeechOptions.builder()
                .model("tts-1")
                .voice(OpenAiAudioApi.SpeechRequest.Voice.NOVA)
                .responseFormat(OpenAiAudioApi.SpeechRequest.AudioResponseFormat.MP3)
                .speed(2.0f)
                .build()));
        byte[] audioBytes = speechResponse.getResult().getOutput();
        Path path = Paths.get("speech-options.mp3");
        Files.write(path, audioBytes);
        return "MP3 saved successfully to " + path.toAbsolutePath();
    }
}
```

spring AI has this speech model

```
AudioController.java  SpeechModel.java x
API. It allows you to convert text input into
lifelike spoken audio.
Since: 1.0.0-M1
Author: Ahmed Yousri
29 @FunctionalInterface
30 public interface SpeechModel extends Model<SpeechPrompt, SpeechResponse> {
31
    Generates spoken audio from the provided
    text message.
    Params: message - the text message to be
    converted to audio
    Returns: the resulting audio bytes
37 default byte[] call(String message) {
38     SpeechPrompt prompt = new SpeechPrompt(message);
39     return call(prompt).getResult().getOutput();
40 }
41
    Sends a speech request to the OpenAI TTS
```

```
AudioController.java  SpeechModel.java  OpenAiAudioSpeechModel.java x
38
    OpenAI audio speech client implementation for
    backed by OpenAiAudioApi.
    Since: 1.0.0-M1
    See Also: OpenAiAudioApi
    Author: Ahmed Yousri, Hyunjoon Choi, Thomas
    Vitale, Jonghoon Park
49 public class OpenAiAudioSpeechModel implements SpeechModel, StreamingSpeechModel {
50
    The speed of the default voice synthesis.
    See Also: OpenAiAudioSpeechOptions
55 private static final Float SPEED = 1.0f;
56
57 private final Logger logger = LoggerFactory.getLogger(getClass());
58
    The default options used for the audio
```

```
14@RestController
15@RequestMapping("/api")
16public class AudioController {
17
18    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;
19    private final SpeechModel speechModel;
20
21    AudioController(OpenAiAudioTranscriptionModel transcriptionModel,
22                    SpeechModel speechModel) {
23        this.openAiAudioTranscriptionModel = transcriptionModel;
24        this.speechModel = speechModel;
25    }
26}
```

```
@GetMapping("/speech")
String speech(@RequestParam("message") String message) throws IOException {
    byte[] audioBytes = speechModel.call(message);
    Path path = Paths.get("output.mp3");
    Files.write(path, audioBytes);
    return "MP3 saved successfully to " + path.toAbsolutePath();
}
```

```
AudioController.java x SpeechModel.java OpenAiAudioSpeechModel.java
25 public class AudioController {
58 }
59
60 @GetMapping("/speech-options")
61 String speechWithOptions(@RequestParam("message") String message) throws IOException {
62     SpeechResponse speechResponse = speechModel.call(new SpeechPrompt(message,
63         OpenAiAudioSpeechOptions.builder().voice(OpenAiAudioApi.SpeechRequest.Voice
64             .speed(2.0f)
65             .responseFormat(OpenAiAudioApi.SpeechRequest.AudioResponseFormat.M
66 Path path = Paths.get("speech-options.mp3");
67 Files.write(path, speechResponse.getResult().getOutput());
68 return "MP3 saved successfully to " + path.toAbsolutePath();
69 }
70
71
```

From prompt to picture - Image generation with ImageModel

Code Sample for image generation

eazy
bytes

```
@RestController
@RequestMapping("/api")
public class ImageController {

    private final ImageModel imageModel;

    ImageController(ImageModel imageModel) {
        this.imageModel = imageModel;
    }

    @GetMapping("/image-options")
    String generateImageWithOptions(@RequestParam("message") String message) {
        var imageResponse = imageModel.call(new ImagePrompt(message,
            OpenAiImageOptions.builder()
                .quality("hd")
                .height(1024)
                .width(1024)
                .style("natural")
                .responseFormat("url")
                .build()));
        return imageResponse.getResult().getOutput().getUrl();
    }
}
```




```
ImageController.java  ImageModel.java x
1  >  /.../
16
17  package org.springframework.ai.image;
18
19  import org.springframework.ai.model.Model;
20
21  @FunctionalInterface
22  public interface ImageModel extends Model<ImagePrompt, ImageResponse> {
23
24      ImageResponse call(ImagePrompt request);
25
26  }
27
```

```
ImageController.java x  ImagePrompt.java  ImageOptions.java  OpenAiImageOptions.java
15  public class ImageController {
24  String generateImage(@RequestParam("message") String message) {
25      var imageResponse = imageModel.call(new ImagePrompt(message));
26      return imageResponse.getResult().getOutput().getUrl();
27  }
28
29  @GetMapping("/image-options")
30  String generateImageWithOptions(@RequestParam("message") String message) {
31      var imageResponse = imageModel.call(new ImagePrompt(message,
32          OpenAiImageOptions.builder()
33              .N(1)
34              .quality("hd")
35              .style("natural")
36              .height(1024)
37              .width(1024).responseFormat("url").build()));
38      return imageResponse.getResult().getOutput().getUrl();
39  }
40
```